OPERATION AND CONFIGURATION MANUAL

Laser Barcode Reader

Model BA-440 DualBeam including Access Control Options

revised 8/1/2009

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FOREWORD

The purpose of this manual is to provide information on how to configure and operate the BA-440 DualBeam Barcode Reader. Barcode Automation Inc. has made every effort to insure that the information in this manual is both accurate and adequate. It is recommended, in the interest of safety and efficiency, that each section be read carefully before installing or servicing this system.

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Communicating with the BA-440 DualBeam

To change the reader configuration or access the onboard diagnostics a computer or laptop is needed that has a serial port or USB port and runs a **terminal emulator** program.

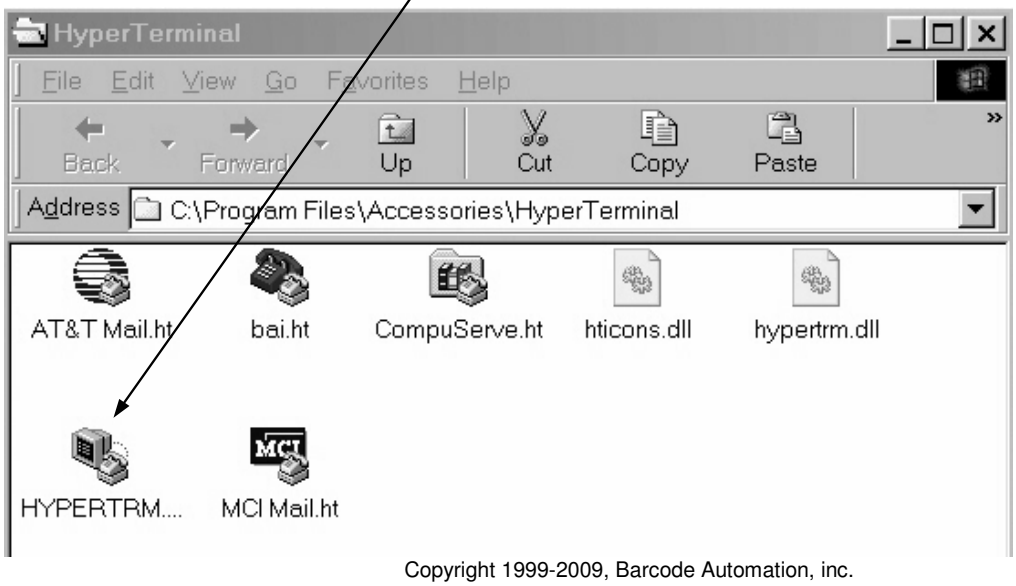
For people using Windows 95/98/2000/ME/XP there is a built in terminal emulator called HyperTerminal that can be set up to communicate with the BA-440 DualBeam. To set up HyperTerminal, follow these steps:

Using Windows HyperTerminal

The Windows HyperTerminal program can be used with a desktop computer or laptop to communicate with the BA-440 DualBeam reader. To set up communications, you will need to:

1. Turn power on to the BA-440 DualBeam.
2. Connect a serial data cable (DB9 male to DB9 female) or a USB cable between the computer and the BA-440 DualBeam.
3. **When using the USB cable you must install drivers first**. You can obtain the drivers from the CD that shipped with the BA-440 DualBeam or online at www.barcode-automation.com.

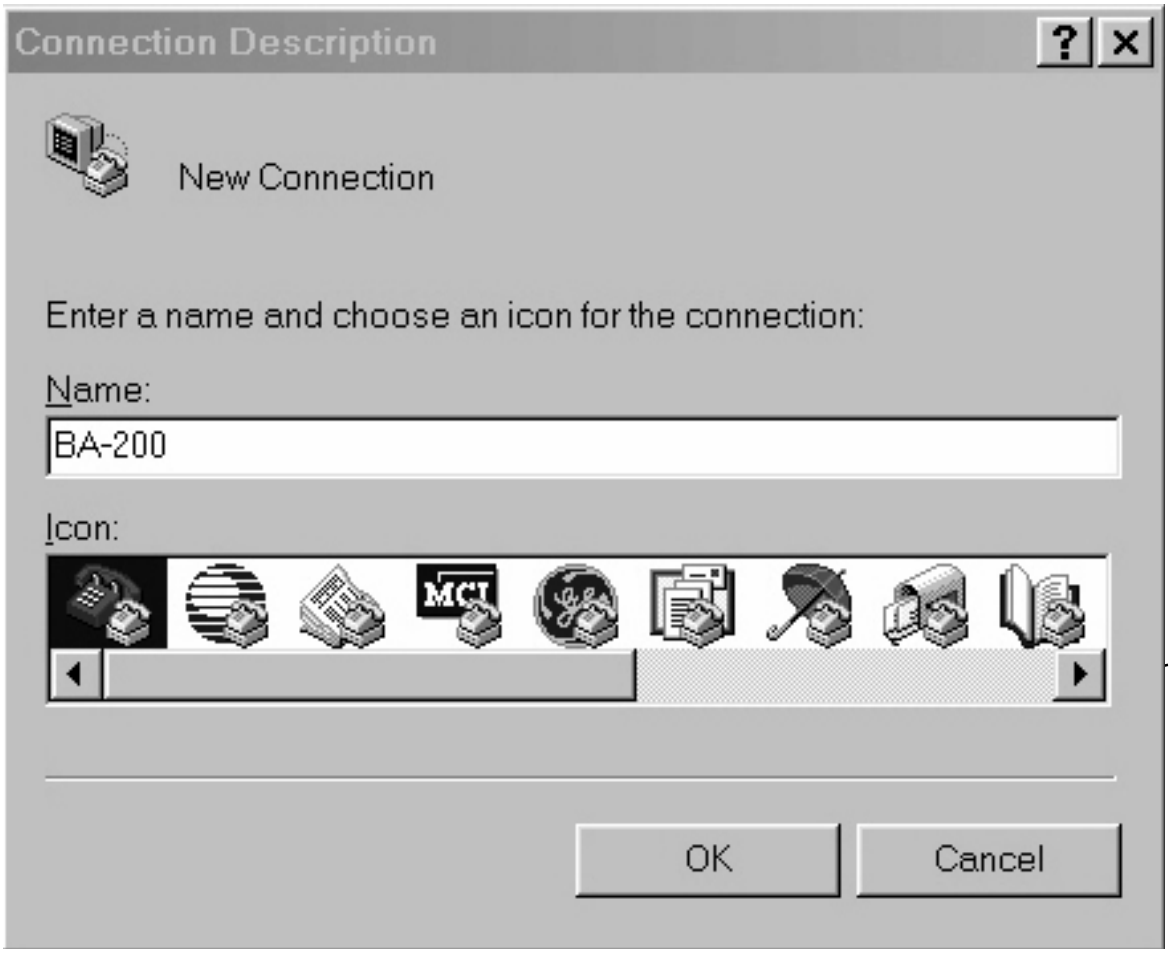
To start HyperTerminal, double click on the Hypertrm.exe icon in the HyperTerminal folder:



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Using Windows HyperTerminal continued…

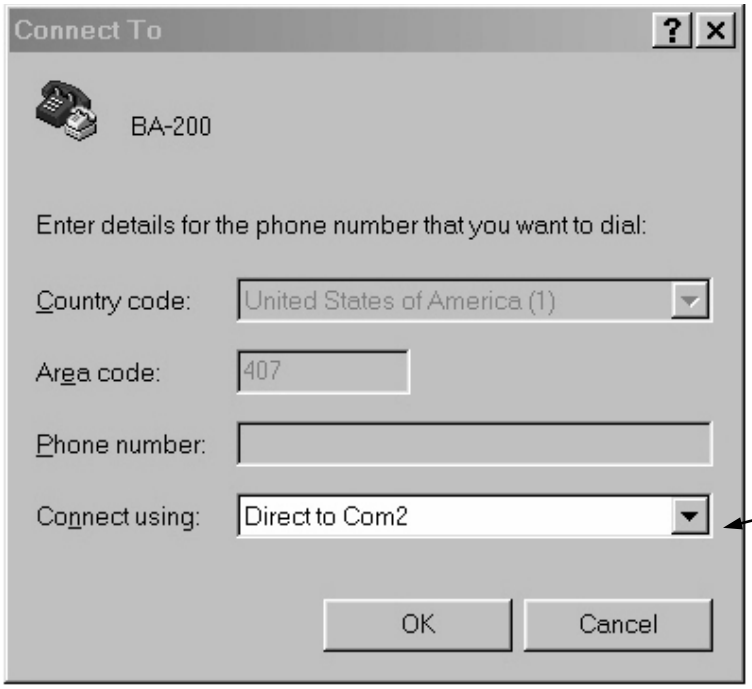
The next screen is:



Enter a name for the new connection so that you can save the session and start it up again easily. You can also choose an icon from those displayed. When done, click on the OK button.

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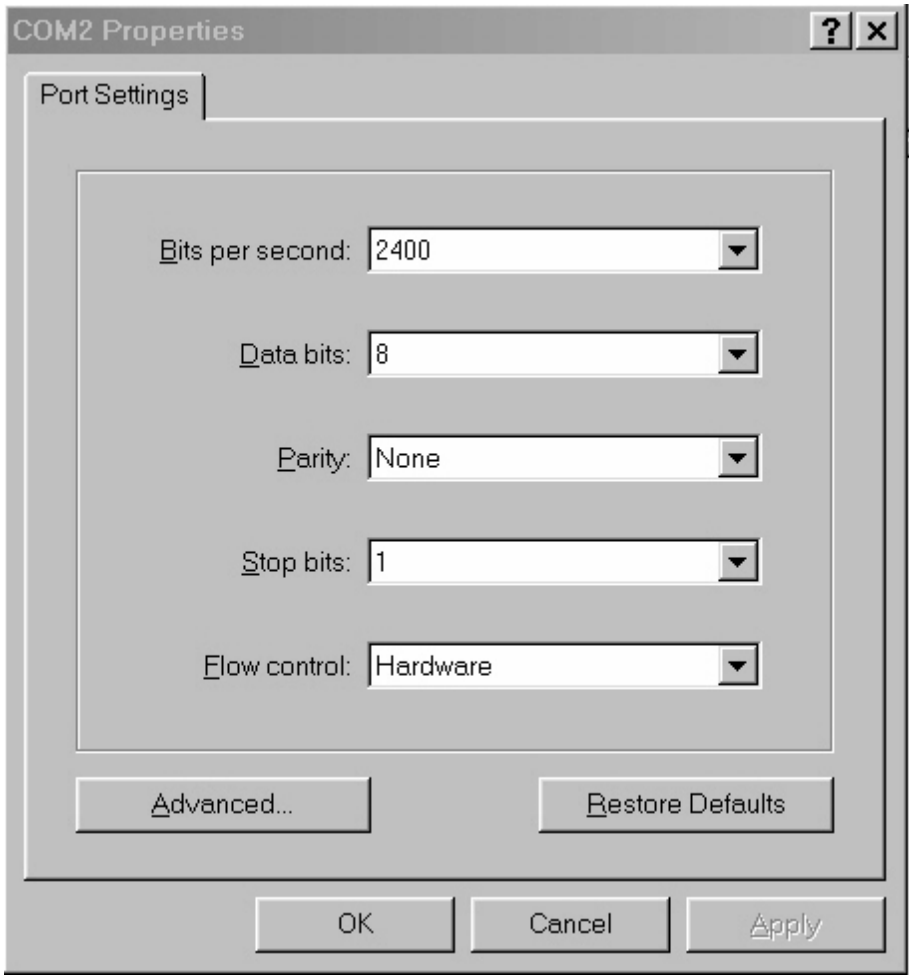
Using Windows HyperTerminal continued…



Select the serial port you will use on the desktop computer or laptop and click OK. **Note:** You must choose a serial port when connecting to the reader with a USB cable.

Do not choose a modem to connect via serial port to the BA-440 DualBeam.

Once that is done the Port Settings menu appears:

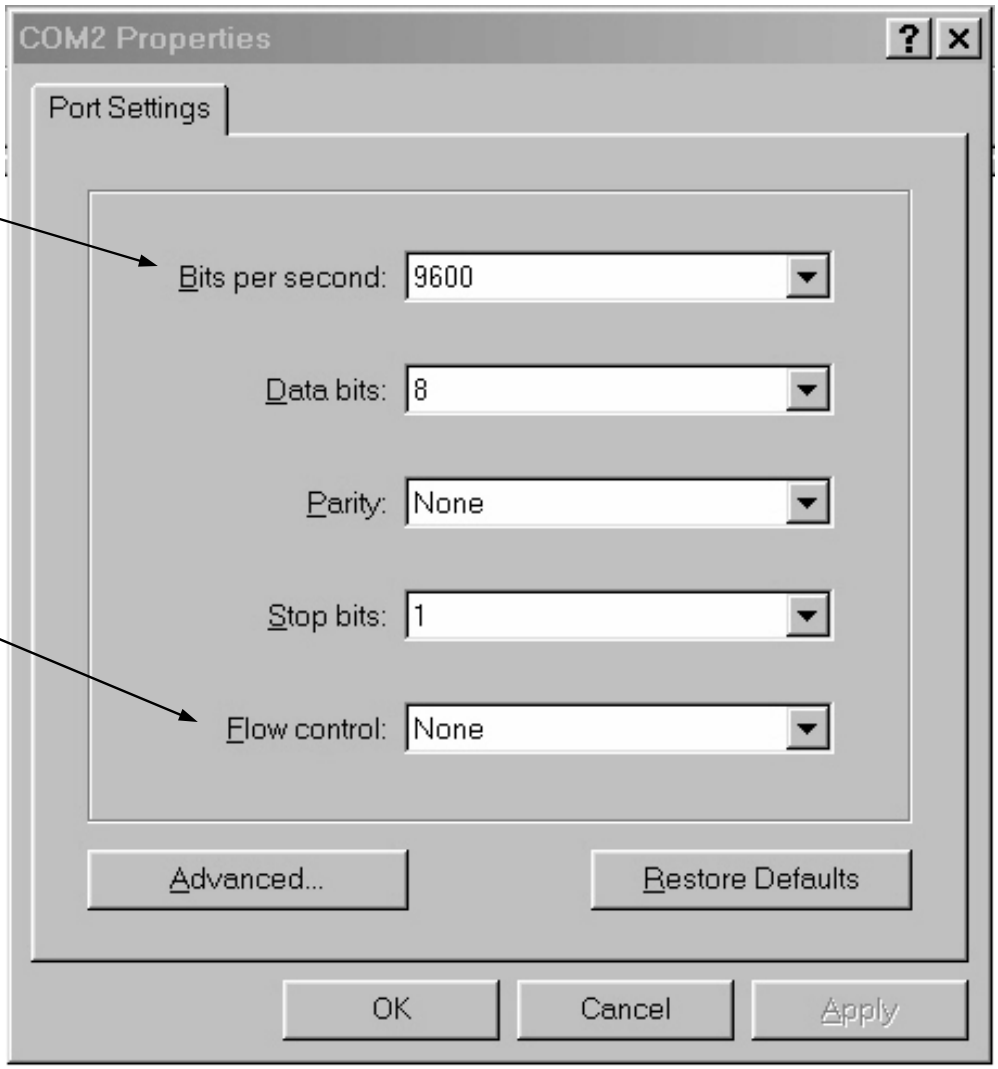


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Using Windows HyperTerminal continued…

Change Bits per second to 9600

Change Flow control to None



Click OK when finished. If the selected serial port on your computer is not available or used by another device the following window will appear:



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Using Windows HyperTerminal continued…

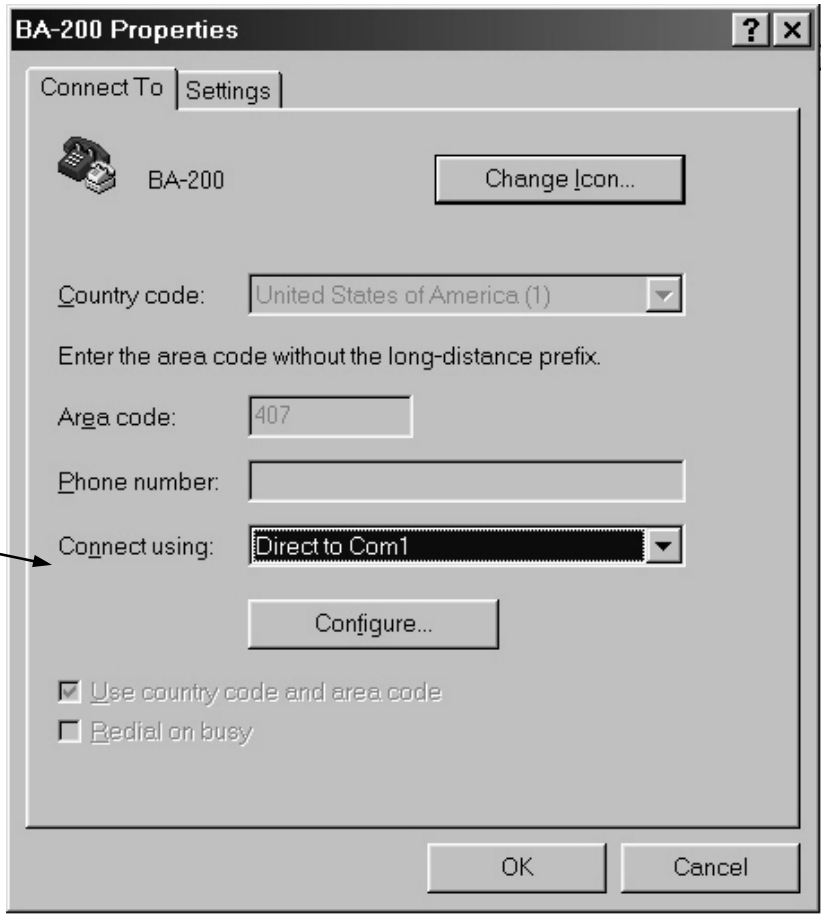
Click OK to clear the window, then click once on the word **File** in the upper left hand corner, then go down and click on **Properties**.



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Using Windows HyperTerminal continued…

Select a different Com port and click OK.



Turn the BA-440 DualBeam reader on. If the communication connection is correct, the power up message is displayed:

#Model BA-440 DualBeam
#Advanced Technology Barcode Reader
#Serial Number XXXXXXX-XX
#Firmware Version 0905291
#RAM Test Passed
#Motor Test Passed

If the message does not appear contact BAI Customer Service.

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How to Set Up the BA-440 DualBeam

The BA-440 DualBeam Reader can be easily configured for many different applications. When communicating with the unit remember these general rules:

1. Commands can be upper or lower case
2. **All** commands end with a carriage return (Enter key)
3. Any changes made are automatically saved - there is no separate save command
4. If a command is not entered properly the reader will respond with an “invalid command” message

Before getting into specifics here is the factory default configuration:

Configuration for BA-440 DualBeam Barcode Reader
Serial Number XXXXXXX-XX
Gain Channel Both
Label Format BAI-10
Vehicle Detect Mode Scan
Wiegand Bits 26
Wiegand Pulse Width 7
Wiegand Pulse Interval 7
Verify Vehicle Number Off
Minimum Reads 1
Communication Baud Rate 9600
Communication Parameters N81
Good Read Message "~F ~L~r~n"
Scan Detect Timeout 1000 ms
Good Read Pulse 1000 ms
Beeper Time 200 ms
Facility Code Shift Off

This configuration is generally used for applications where the BA-440 DualBeam will read decals and transmit them to a security panel or telephone entry system through the 26 bit Wiegand communication port. If you have any questions contact BAI at 800-528-9167.

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How to Set Up the BA-440 DualBeam

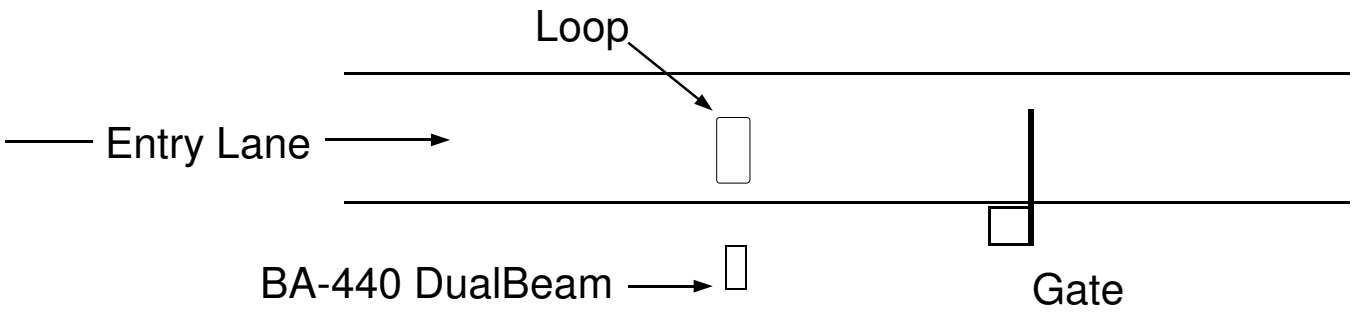
Setting the Label Format

In order to read barcode decals properly the BA-440 DualBeam must be programmed for the correct format. Installations at new sites use the BAI-10 format but when replacing an existing barcode scanner you **may** need to configure the BA-440 DualBeam to read the older BAI-8 or CB-6 label format. Note: the BA-440 DualBeam **will not** read the AFC label format. Contact BAI at 800-528-9167 for assistance in configuring your Reader.

Using an Arming Trigger with the BA-440 DualBeam

If desired, an arming signal can be connected to the BA-440 DualBeam through the Aux In- & Aux In+ terminals. This signal will control when the reader will attempt to read vehicles, and can also control when the laser is on. Using an external trigger, it is possible to turn the laser off until the Aux In signal is detected by the reader. **Note: There is no requirement to have a Aux input.** The BA-440 DualBeam will operate fine without one.

In this example, a Loop detector output is wired to the Aux input of the reader. Usually the loop will be located directly in front of the reader or just ahead of it as shown:



The loop detector connection to the BA-440 DualBeam must be the Normally Open (NO) type. When the detector senses a vehicle is present the connection should close.

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How to Set Up the BA-440 DualBeam

Activating the Aux Input

In addition to connecting the arming trigger to the Aux In- & Aux In+ terminals the Vehicle Detect Mode parameter must be changed from **Scan** to **Loop** by typing **vdm loop** and press **Enter**. Some new parameters will now appear in the configuration:

Laser Diode Saver Off - controls when the laser is on. When set to OFF, the laser will be on constantly. If set to ON, the laser will only turn on when the Aux In input is active. To change the setting type **lds on** or **lds off** and press **Enter**.

Loop Delay Time 100 ms - this is a programmable delay that extends the Aux In signal "on" time after the vehicle has cleared the loop. It is useful when the arming loop is located ahead of the reader.

No Read Pulse 1000 ms - since the Aux In input lets the reader know when a vehicle is present, the BA-440 DualBeam can now detect whether or not a barcode was read. When a vehicle passes that does not have a barcode the reader will close the Aux Out relay for 1000 ms (1 second), which can be connected to an alarm or indicator.

Wiegand No Read On - if a vehicle passes by without a barcode the reader can send a special message out the Wiegand port to alert the security panel or telephone entry system that no read has occurred. This message is all zero's and will be sent only if this parameter is set ON. To change this setting, type **wnr on** or **wnr off** and press **Enter**.

If you have any questions about using a Loop or other vehicle detection device with the BA-440 DualBeam contact BAI at 800-528-9167.

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How to Set Up the BA-440 DualBeam

Activating Standalone operation in the BA-440 DualBeam

Standalone mode allows the BA-440 DualBeam to control one gate directly. The gate control relay output is a dry contact at the GdRead- & GdRead+ terminals. **NOTE: The only decal formats that are supported in standalone mode are the BAI-8 and BAI-10.** If you are using CB-6 decals this mode **will not function**.

1. Start standalone mode by typing **vvn on** and pressing **Enter**.
2. Next, enter the Facility Code for the decals by typing **fc XXX** and pressing **Enter**. (XXX is the decal Facility Code).
3. Set the Lowest Vehicle Number to be accepted by typing **lvn XXXXX** and pressing **Enter**. (**XXXXX** is a number) You do not need to use leading zero's in the number.
4. Set the Highest Vehicle Number to be used by typing **hvn XXXXX** and pressing **Enter**. (**XXXXX** is a number) You do not need to use leading zero's.
5. Check the Access Table by typing **a?** and pressing **Enter**. The screen will display the number range and access status. Be sure that access is granted for all the decals you want to enter.
6. Pass a valid decal by the reader. The audible signal sound beep for approx 1/4 second if the decal was accepted.
7. To delete a number from the Access Table, type **ad XXXXX** and press **Enter**. The **XXXXX** is the ID number.
8. To Grant Access to an ID number type **ag XXXXX** and press **Enter**. **XXXXX** is the decal ID number.

NOTE: You can also grant or deny access in blocks. The commands will be **ad XXXXX - XXXXX** or
ag XXXXX - XXXXX and press **Enter.**

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Test Mode for Installation Assistance

The BA-440 DualBeam has a **test** mode that transmits an internally generated test barcode. This mode is useful for verifying that Wiegand communications connections are correct and the reader electronics are operating properly. To access **test** mode, you will need to:

1. Turn the Reader on
2. Connect using a laptop computer running a terminal program
3. Type the command **test** and press the Enter key. NOTE: If Vehicle Detect Mode is set to LOOP, the Aux In signal must be active (on) before the reader will operate. The reader will respond with:

test
Test Mode On

barcode on
Test mode 123 12345 14
Test mode 123 12345 14
Test mode 123 12345 14
Test mode 123 12345 14

These messages indicate that the reader is decoding and transmitting test barcode number 123 12345 about once every second. On the Wiegand port, the facility code is transmitted as 123 and the vehicle ID code is 12345. You can check to see if this is the data received on the other end of the Wiegand connection to verify that communications are wired correctly.

To exit **test** mode, type **test** and press the Enter key again or simply turn the barcode reader off. When you turn it back on it will boot up in regular operating mode.

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Configuration Commands

When using these commands, remember these points:
1. Commands may be entered in upper or lower case.
2. To execute a command always hit the <**Enter**> key.
3. The factory default option is shown in brackets [BAI-10].

Configuration

Command: **CFG**

Purpose: Displays the current configuration of the BA-440 DualBeam.

Options: none

Example: **CFG**

Result: **Note: The actual configuration may be different depending on how the BA-440 DualBeam is set.**

Configuration for BA-440 DualBeam Barcode Reader
Serial Number XXXXXXX-XX
Gain Channel Both
Label Format BAI-10
Vehicle Detect Mode Scan
Wiegand Bits 26
Wiegand Pulse Width 7
Wiegand Pulse Interval 7
Verify Vehicle Number Off
Minimum Reads 1
Communication Baud Rate 9600
Communication Parameters N81
Good Read Message "~F ~L~r~n"
Scan Detect Timeout 1000 ms
Good Read Pulse 1000 ms
Beeper Time 200 ms
Facility Code Shift Off

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Configuration Commands

Send Configuration

Command: **SEND CFG**

Purpose: transmits the current configuration data to the terminal in a format that can be saved and uploaded back into the reader.

Options: none

Example: **SEND CFG**

Result: **Note: The actual configuration may be different depending on how the BA-440 DualBeam is set.**

#Configuration for BA-440 DualBeam Barcode Reader
#Serial Number 0903001-AA
GC Both
LF BAI-10
VDM Scan
WB 26
WPW 7
WPI 7
VVN Off
MR 1
CBR 9600
CP N81
GRM "~F ~L~r~n"
SDT 1000 ms
BT 200 ms
FCS Off

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Barcode Label Format Commands

Label Format

Command: **LF**

Purpose: Set the barcode format to be read. The following formats are available:

BAI-10 - 10 digit barcode used for new installations.

BAI-8 - 8 digit barcode used in many older installations.

CB-6 - 6 digit barcode used by Accu-Sort Model Auto/ID-24 scanners. Select this only when replacing Accu-Sort scanners using the CB-6 format.

Options: [BAI-10], BAI-8, CB-6

Example: **LF BAI-8**

Note: the BA-440 DualBeam does not support the old AFC decal format.

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Barcode Label Format Commands

Facility Code Digits

Command: **FCD** used in BAI-8 Label Format only.

Purpose: When reading BAI-8 barcodes, this sets the number of digits on the decal that are used for the facility code. It is necessary to select how many of the 8 digits are used for the facility code.

Options: [2], 3

Example: **FCD 3**

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Barcode Label Format Commands

Site Code

Command: **SC**

Purpose: Used in CB-6 label format only to set the site code that is transmitted out the Wiegand port.

Options: [00]-4095

Example: **SC 57**

Facility Code Shift

Command: **FCS**

Purpose: Used in BAI-10 label format only to change a facility code value on the barcode to a valid number that will be transmitted out the Wiegand port. Up to 8 different facility code values may be shifted to numbers between 0 and 255.

Options: ON, [OFF] FCS xxx>yyy

First turn the function on by typing **FCS ON** and hitting <enter>, then type **FCS**, a space, the barcode facility code (xxx), a "**>**" sign, the desired facility code number for transmission (1-255), then the <enter> key.

FCS 368>019
If a barcode with a facility code of 368 is read, the BA-440 will shift the facility code value to 019 and transmit the data out all communication ports.

FCS 368>019 359>019
A barcode with facility code 368 or 359 will have the number shifted to 019 and transmitted out all communication ports.

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Read Trigger Commands

Vehicle Detect Mode

Command: **VDM**

Purpose: Selects when the BA-440 DualBeam will read decals.

Scan - The reader looks continuously for barcodes.

Loop - This mode requires a loop detector or other vehicle presence detector connected to the Aux In terminals to activate the reader. When the input signal is off, the reader is disabled. When the signal is on, the BA-440 DualBeam will look for and read barcodes, transmitting data as soon as they are decoded. In addition, if the input signal cycles on and off but no barcodes were read the BA-440 DualBeam will transmit a “no read” message to show that no barcodes were read.

Options: [Scan], Loop

Example: **VDM LOOP**

Loop Delay Time

Command: **LDT**

Purpose: Used in loop mode only, this sets how long the reader will continue looking for a decal after the Aux Input goes Off.

Options: any number from 0 to 2500 in multiples of 10 mS. [100] 1000 = 1 second.

Example: **LDT 1000**

Note: This command disabled if Vehicle Detect Mode is Scan.

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Read Trigger Commands

Laser Diode Saver

Command: **LDS**

Purpose: Used in loop detect mode only, this turns off the laser when the Aux input signal from the vehicle presence detector is off.

Options: on, [off]

Example: **LDS Off**

Note: This command disabled if Vehicle Detect Mode is Scan.

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Wiegand Communication Commands

Wiegand Bits

Command: **WB**

Purpose: Set Wiegand bit format. Only use 26 bit Wiegand unless advised to change the setting by BAI.

Options: off, [26], 30, 31

Example: **WB 26**

Wiegand No Read

Command: **WNR**

Purpose: When the Vehicle Detect Mode is set to Loop, this command allows the BA-440 DualBeam to send all zero's out the wiegand port if a valid barcode is not read.

Note: This command is disabled when Wiegand Bits is off.

Options: [on], off

Example: **WNR OFF**

Wiegand Pulse Width

Command: **WPW**

Purpose: Set the duration (width) of the Wiegand Pulses. Do not change unless advised by BAI.

Options: [1 - 15]

Example: **WPW 7**

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Wiegand Communication Commands

Wiegand Pulse Interval

Command: **WPI**

Purpose: Set the time interval between Wiegand Pulses. Do not change unless advised by BAI.

Options: [5 - 15]

Example: **WPI 7**

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Standalone Access Control Commands

Verify Vehicle Number

Command: **VVN** **Note:** All options except REM are disabled when using the CB-6 label format.

Purpose: Enable or disable the standalone access control function of the BA-440 DualBeam. Available options are:

Off - “standalone” access control is disabled.
On - the reader will use the internal database to verify vehicle numbers and grant or deny access.
REM - Remote mode. For more information, see the section on Remote Commands.
DAR - Delete After Read. Decals are allowed access only once.

Options: on, [off], REM, DAR

Example: **VVN ON**

Facility Code

Command: **FC**

Purpose: Set the facility code for a particular location. A facility code is a number associated with the gated community, business, or parking area where the reader is installed. All barcodes that are valid for this location must have this facility code. Command is disabled when Verify Vehicle Number is Off. Only one Facility code can be set.

Options: [0]-999

Example: **FC 21**

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Standalone Access Control Commands

Lowest Vehicle Number

Command: **LVN**

Purpose: Set lowest allowable vehicle ID number in the database. If this number is higher than the Highest Vehicle Number, the Highest Vehicle Number will be automatically changed to this number. This command is disabled when Verify Vehicle Number is off.

Options: 00000 to 99999

Example: **LVN 1**

Highest Vehicle Number

Command: **HVN**

Purpose: Set highest allowable vehicle ID number in the database. If this number is lower than the Lowest Vehicle Number, the Lowest Vehicle Number will be automatically changed to this number. This command disabled when Verify Vehicle Number is off.

Options: 00000 to 99999

Example: **HVN 60000**

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Standalone Access Control Commands

Access ?

Command: **A?**

Purpose: Show access rights for all ID numbers in the database.

Options: A? or A? XXXXX

Example: **A?** will display the access status of all numbers.
A? XXXXX will display access status of an individual number.

Access Grant

Command: **AG**

Purpose: Grant access rights for a single vehicle ID number or range of numbers.

Options: Any number from lowest vehicle number to highest vehicle number.

Example: **AG 51000** (or block load as “ag 51000 - 51200”)

Access Deny

Command: **AD**

Purpose: Deny access rights for a single vehicle ID number or range of numbers.

Options: Any number from lowest vehicle number to highest vehicle number.

Example: **AD 51000** (or block deny as “ad 51000 - 51200”)

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Serial Communication Commands

Communication Baud Rate

Command: **CBR**

Purpose: Set the communication speed (baud rate) for the serial port.
Note: Once the parameter has been changed, it will not take effect until the BA-440 DualBeam is restarted.

Options: 2400, 4800, [9600], 19200, 38400

Example: **CBR 9600**

Good Read Message

Command: **GRM "Text ~F ~L~r~n"**

Purpose: Set up the format of the message sent out the RS232 serial port and USB port for each good read. Up to 18 characters can be selected and must be enclosed in quotes "" along with any special designators desired.

Options: Up to 18 characters including special designators
"~F" - this will print the facility code from the decal
"~L" - this will print the decal ID number
"~C" - this will print the Check digit (only if a check digit is used)
"~T" - this will print the time stamp for the message from the BA-440 DualBeam internal clock
"~r" - this will print a carriage return
"~n" - this will print a line feed

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Serial Communication Commands

Example: **GRM “Read ~F ~L ~C~r~n”**
When the BA-440 DualBeam reads barcode 0001234550 the message from the serial port will be:

Read 000 12345 50

Example 2: **GRM “Reader 1 ~L~r~n”**
When the BA-440 DualBeam reads barcode 00123457 the message from the serial port will be:

Reader 1 12345

No Read Message

Command: **NRM "Text ~F~L~C~r~n"**

Purpose: Set up the format of the message sent out the RS232 serial port and USB Port for each no read. This command is disabled if Vehicle detect mode is Scan. Up to 18 characters can be selected and must be enclosed in quotes ”” along with any special designators desired.

Options: Up to 18 characters including special designators
“~F” - this will print the facility code from the decal
“~L” - this will print the decal ID number
“~C” - this will print the Check digit (only if a check digit is used)
“~T” - this will print the time stamp for the message from the BA-440 DualBeam internal clock
“~r” - this will print a carriage return
“~n” - this will print a line feed

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Serial Communication Commands

Example: **NRM “No Read ~r~n”**

When the BA-440 DualBeam has a no read the message from the serial port will be:

No Read

Example 2: **NRM “Reader No Read ~T~r~n”**

When the BA-440 DualBeam has a no read the message from the serial port will be:

Reader No Read 10/24/06 13:38:48.1

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Input/Output Commands

Good Read Pulse

Command: **GRP**

Purpose: Sets the duration of the good read relay output in milleseconds. This pulse is generated when a valid label is read.

Options: 10 to 2500 (in multiples of 10 ms) [1000]

Example: **GRP 500**

No Read Pulse

Command: **NRP**

Purpose: Sets the duration of the no read relay output in milleseconds. This pulse is generated under only two conditions:

When Vehicle Detect Mode is set to Loop and no decal is read, the No Read Pulse will activate.

When Verify Vehicle Number is On and access is denied to a particular barcode ID.

Options: 10 to 2500 (in multiples of 10 ms) [1000]

Example: **NRP 1000**

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Input/Output Commands

Scan Detect Timeout

Command: **SDT**

Purpose: Sets the amount of time in milleseconds that the BA-440 DualBeam must wait (without reading a decal) before it will read the **same** decal again. In practice, if a barcode is placed on the laser line the reader will decode it and transmit data once. In order for the BA-440 DualBeam to read and transmit **that particular barcode again** it must be moved out of the laser scan line for **at least** the length of the scan detect timeout setting. If the setting is 1000, then the barcode must be out of the laser line for at least 1 second before the reader will decode and transmit it again. If the reader sees a different label, however, it will read it immediately regardless of the scan detect timeout value.

Options: Any number from 10 to 2500 (in multiples of 10 ms) [1000]

Example: **SDT 1000**

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Miscellaneous Commands

Gain Control

Command: **GC**

Purpose: Determines what receiver channels are used to decode barcodes. The BA-440 DualBeam processes the barcode data using two channels, one for Lo gain and one for Hi gain. New, bright decals have a strong signal and are processed successfully on the Lo gain channel. Older, less reflective decals can have a very weak signal that requires processing on the Hi gain channel to be read.

Options: [Both], Lo, Hi

Example: **GC Lo**

Be careful when changing this parameter. Setting the Gain Control to the Lo channel only may prevent older or damaged decals from reading while still picking up newer ones. Setting the Gain Control to the Hi channel only may prevent newer decals from reading while picking up older ones. In general this parameter is used to troubleshoot problems and should not be changed without direction from BAI.

If the setting is changed the technician will need to observe a large number of vehicles passing the reader to determine if barcodes are being read successfully.

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Miscellaneous Commands

Minimum Reads

Command: **MR**

Purpose: Set the minimum number of times the BA-440 DualBeam must read a decal before considering it a valid barcode. If set to 2, the reader must scan and decode a decal twice with the exact same number for it to be valid.

Options: [1], 2, 3

Example: **MR 2**

Echo

Command: **ECHO**

Purpose: Used to enable or disable the echoing of keyboard characters to the screen.

Options: [on], off

Example: **ECHO ON**

Time

Command: **TIME**

Purpose: Show current time and date on the BA-440 DualBeam internal clock.

Options: none

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Miscellaneous Commands

Set Date

Command: **SET DATE MM/DD/YY**

Purpose: Set date for the BA-440 DualBeam internal clock.

Options: none

Example: **SET DATE 09/07/06**

09/07/06 12:58:19

Set Time

Command: **SET TIME HH:MM:SS**

Purpose: Set time for the BA-440 DualBeam internal clock.

Options: none

Example: **SET TIME 08:22:00**

09/07/06 08:22:00

Beeper Time

Command: **BT**

Purpose: Sets the duration of the Read beep in milleseconds. If set to zero, the beeper is disabled.

Options: 10 to 2500 (in multiples of 10 ms) [200]

Example: **BT 1000**

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Diagnostic Commands

Transaction Log Status

Command: **TLS**

Purpose: Show the status of the transaction log. Includes total number of transactions, and transactions lost (due to buffer overflow) since the last erase transaction log command. **Total Transactions** is the number of transactions since the last Erase Transaction Log command. **Lost Transactions** is the number of old transactions that have passed out of the transaction log (it only stores the last 2000).

Options: none

Example: **TLS**
Transaction Log Status: as of 11/20/06 16:53:56.6
Total Transactions = 44049
Lost Transactions = 42049

Transaction Log

Command: **TL** or **TL (1 - 2000)**

Purpose: Displays transactions stored in the log. The BA-440 DualBeam stores the last 2000 transactions including the time and date. This command will dump selected entries out the serial port and USB port so that they can be captured by the technician. If only the command **tl** is entered all of the log file entries will be sent out the serial port. By adding a qualifier after the **tl** command you can specify how many entries you wish to see.

Options: none

Example: **TL 20**

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Diagnostic Commands

Erase Transaction Log

Command: **ERASE TRANSACTION LOG**

Purpose: Clears the transaction log and zeros the transaction log counters.

Options: none

Example: **ERASE TRANSACTION LOG**

Status

Command: **STATUS**

Purpose: Displays current temperature information in the reader

Options: none

Example: **STATUS**

PCB= 73F Motor=22ms

Displays temperature inside enclosure and motor speed

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Diagnostic Commands

Read Rate

Command: **RR**

Purpose: Displays current read rate for the reader. To run this test, you will need to hold a barcode steady in the laser line. To stop this diagnostic, type “rr” and hit the enter key.

Options: none

Example: **RR**

```
Diagnostic Mode On
Test Read Lo=98/0 Hi=98/0 PCB=76F Motor=19ms
Test Read Lo=99/1 Hi=100/0 PCB=76F Motor=19ms
Test Read Lo=100/0 Hi=99/0 PCB=76F Motor=19ms
Test Read Lo=99/0 Hi=100/0 PCB=76F Motor=19ms
```

Test Read indicates the read rates on the low gain and high gain channels of the decoder. Most decals will only show 99 or 100 on only one channel at a time.

PCB is the current temperature inside the reader.

Motor indicates motor speed.

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Diagnostic Commands

Hours

Command: **HOURS**

Purpose: Displays hours of operation and temperature history.

Options: none

Example: **HOURS**

```
Hours on       1286
Laser on       1286
PCB   < 50F    0
PCB   50-70F   69
PCB   70-90F   1201
PCB   90-110F  16
PCB   110-130F 0
PCB   > 130F   0
```

Log

Command: **LOG**

Purpose: Sends all onboard diagnostic data for the reader out the serial and USB ports to be captured and analyzed by service technicians.

Options: none

For assistance with this command contact BAI Customer Service at 800-528-9167.

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Diagnostic Commands

Diagnostic Log

Command: **DLOG**

Purpose: Displays the diagnostic log of temperature information in the reader for the last 255 hours.

Options: none

Example: **DLOG**

Hours on 2077
Laser on 2077
PCB < 50F 0
PCB 50-70F 0
PCB 70-90F 1305
PCB 90-110F 229
PCB 110-130F 530
PCB > 130F 13
-255 hrs. pcb 85F motor 21
-254 hrs. pcb 85F motor 22

* through*

-3 hrs. pcb 83F motor 21
-2 hrs. pcb 83F motor 22
-1 hrs. pcb 73F motor 22

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